

Extending the Early-Session Sentiment Factor in China A-Shares: Six Falsified Directions and a Reusable Falsification Protocol

Chenyang Sun

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Abstract

This report documents an attempt to extend a previously validated A-share timing strategy, the *early-session sentiment factor*, in which the cross-sectional fraction of stocks trading below the open at 09:55 predicts intraday continuation. Two extension lines were pursued: an *inward* line, in which the information content of the morning move was re-measured (cross-sectional dispersion, trade-amount-weighted breadth, intra-window path shape), and an *outward* line, in which the compression-window recipe was transplanted to new windows and new targets (tail-of-day breadth, post-holiday reopenings, realized-variance timing). All tests were conducted under an execution-honest protocol with point-in-time universes, survivorship-free data, one-shot out-of-sample validation, multi-tier transaction costs, and family-wide multiple-testing control over 25 preregistered trials. None of the six directions yielded independently deployable alpha. One proposition was nevertheless confirmed: an information window orthogonal to the morning session exists (tail breadth predicts the overnight gap with $t = 3.01$ at a correlation of 0.106 with the parent signal), although its payoff cannot be monetized. The principal contribution is therefore methodological: a reusable falsification toolkit comprising a level-orthogonality screen, nested HAC incremental regressions, a sign-flip rejection rule for one-shot out-of-sample tests, permutation inference for small-sample calendar regimes, a HAR-baseline requirement for volatility claims, a run-to-net-value requirement, a registry of microstructure artifacts, and a two-layer verification process by which three apparent discoveries were rejected before acceptance. It is argued that a disciplined chain of negative results, together with one confirmed proposition and reusable infrastructure, constitutes the appropriate outcome of the study.

1 Motivation

The parent strategy conditions intraday exposure on a single quantity observable at 09:55: the fraction of investable stocks trading below their open, hereafter *morning breadth*. The strategy has been validated on a wide, survivorship-free cross-section, and its single-factor refinement is exhausted: parameter tuning, threshold ensembles, and hedged variants collapse to noise ($t \approx 0.2$) once multiple testing is controlled. Further value can therefore arise only from new information, orthogonal factors, or transplantation of the underlying mechanism.

That mechanism may be summarized as a compression-window principle:

information-compression window $\times$ cross-sectional breadth $\longrightarrow$ prediction of the subsequent interval,

where the predictive content resides in breadth, interpreted as the distinction between systematic and idiosyncratic participation, rather than in the index-level move itself. Two extension lines follow directly. Under **Line A (inward)**, richer descriptions of the same 09:55 cross-section (dispersion,

trade-amount weighting, intra-window path shape) are examined for incremental information. Under **Line B (outward)**, other compression windows (the close, post-holiday reopenings) and other targets (realized variance) are examined as candidates for replicating the recipe while remaining orthogonal to the parent.

Six mechanism-motivated directions were preregistered and executed (A1–A3 inward; B1, B3, B5 outward). All six were rejected as sources of deployable alpha. The attempts are summarized briefly in Section 3; the findings that survive them, which constitute the core of this report, are presented in Section 4.

2 Research protocol

A single protocol, fixed before any test was run, was applied to all directions.

Universe and data. Main board, ChiNext, and STAR listings, excluding the Beijing exchange; ST names, suspensions, one-sided limit days, and IPOs younger than 250 trading days were excluded point-in-time; delisted stocks were retained (survivorship-free). The sample covers 2015–2025 with approximately 3,600 investable names per day. Minute-level snapshots were obtained from a licensed vendor and frozen to parquet; every derived cache was reconciled 1:1 by row count against the parent price cache.

Execution honesty. Signals were restricted to prices observable at or before 09:55 (lag = 0); fills were repriced to 10:00, and the 15:00 fill corresponds to the closing auction. Equal-weighted open-to-close “intraday legs” were excluded from all performance figures for the reasons given in Section 4. Costs were charged on every position change and stress-tested at 3, 6, and 10 bps-per-side tiers.

Statistical discipline. The in-sample period (IS, 2015–2020) was used for direction and thresholds; the out-of-sample period (OOS, 2021–2025) was evaluated once and never revisited. Inference relied on Newey–West/HAC t -statistics, with permutation tests substituted wherever event counts were small. A project-level trial ledger was accumulated across directions (final count: 25) and used for family-wide Bonferroni control. Only preregistered variants were admitted; no specification search was performed.

Process. Each direction was executed by a researcher required to freeze all reported numbers to interpreter-produced files. Acceptance was conditional on independent re-execution of the headline results from the frozen artifacts by the project lead (“two-layer verification”). A signed, append-only handoff ledger recorded scope, verified claims, assumptions, and unverified risks at every stage.

3 The six directions in brief

Table 1 summarizes the six attempts; the frozen evidence is shown in Figures 1–4. Each direction is compressed below into motivation, method, and verdict.

3.1 Line A (inward): re-measuring the morning cross-section

A1 Dispersion. *Motivation:* a systematic sell-off (uniform decline, low dispersion) should continue, whereas an idiosyncratic one should not. *Method:* the cross-sectional standard deviation of open-to-09:55 returns was tested in nested HAC regressions against breadth and level, in conditional double sorts, and in two preregistered overlay variants on three sleeves. *Verdict:* *null*. Crash mornings in the A-share sample are high-dispersion events, reflecting heterogeneous panic selling,

Table 1: Six preregistered directions, all rejected as deployable alpha. Every headline number was independently reproduced in frozen artifacts before acceptance.

Dir.	Hypothesis (compressed)	Headline IS evidence	OOS / robustness	Verdict and failure
A1	dispersion separates systematic from idiosyncratic sell-offs	interaction $t = -0.40$; corr(disp, target) ≈ 0	$\Delta\text{Sharpe} -0.16$ to -1.16 (6/6 neg.)	Null. Crash more <i>high</i> -dispersion; dispersion volatility.
A2	amount-weighted breadth measures true selling pressure	breadth absorbs all: $t = -4.08$ vs. amount $t = +1.39$	positive 1/6 (noise); volume-weighted 0/6	Null. corr(amt _{xs} , level) = +0.52; level contamination information.
A3	intra-window path shape (acceleration, curvature)	accel. $t = -4.45$ (double-controlled); curvature $t = -0.09$	OOS $t = +0.45$: sign flip	Null. IS significance a level-timing artifact OOS.
B1	tail breadth (14:45–14:55) predicts the overnight gap	IC $t = 3.01$ (full); corr with parent signal 0.106	overlay fails at 10 bps; OOS IC $t = 1.01$	Proposition confirmed alpha. Payoff concentrated overnight leg, structurally $\approx -$
B3	post-holiday reopenings are the strongest compression regime	interaction -312 bps (perm. $p = .046$, $n = 38$)	OOS -60 bps ($p = .32$); long-gap ≈ 0 ($p = .45$)	Null; not independent. Mechanism gradient reversed; corr 0.98.
B5	morning dispersion predicts realized variance (2nd moment)	HAR increment $t = 6-9$; OOS DM -4.16 (survives Bonferroni $\times 25$)	vol-targeting $\Delta\text{Sharpe}(\text{C}-\text{B}) \approx 0$; $\text{C}-\text{A} = -0.12/-0.22$	Null as a strategy. Genuine statistical signal with no commission value; HAR spans the sure choice.

Costs: 1.2 bps commission + 10 bps stamp duty at baseline, stress-tested at 3/6/10 bps tiers. IS 2015–2020; OOS 2021–2025, evaluated in frozen artifacts. “Double-controlled” regressions include both raw breadth and the index-level morning move (HAC, 5 lags).

so the identification “systematic = low dispersion” is not supported by the data. A placebo test shows dispersion amplifying returns symmetrically in both directions, consistent with a volatility proxy rather than a directional signal. All six overlay cells underperform the parent (Figure 1, left panel).

A2 Amount-weighted breadth. *Motivation:* weighting breadth by traded amount was expected to measure genuine selling pressure more accurately than one-stock-one-vote counting. *Method:* morning cumulative trade amounts were used as weights inside breadth; a purified tilt component (amt_{xs}) was constructed, and head-to-head absorption regressions were run. *Verdict:* null. The amount tilt is level contamination: corr(amt_{xs}, level) = +0.52, because high-volume morning declines concentrate in large, liquid names that tend to stabilize intraday. Raw breadth absorbs the entire effect ($t = -4.08$ against +1.39), and the overlays lose in 11 of 12 cells (Figure 1, middle panel).

A3 Intra-window path shape. *Motivation:* two mornings with identical 09:55 breadth may differ in the trajectory by which that level was reached (accelerating decline versus fading pressure),

Line A on the small-cap sleeve: no re-measurement of the morning cross-section improves on the parent overlay

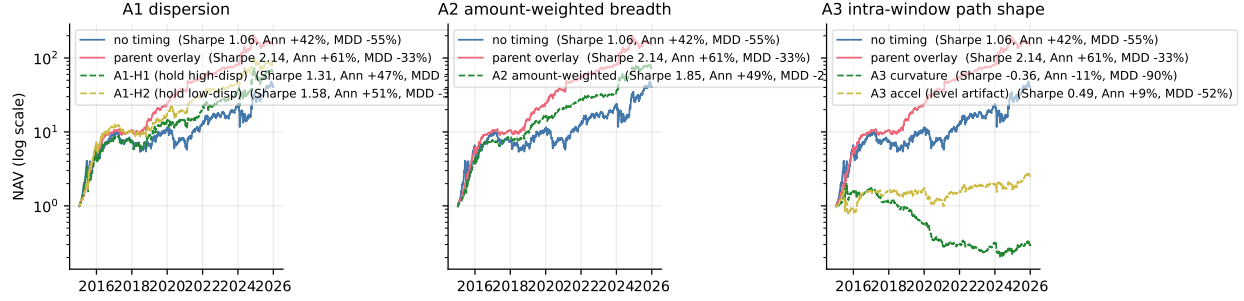


Figure 1: Line A on the small-cap sleeve, 2015–2025, execution-honest (10:00 fills, 1.2 bps + 10 bps costs), logarithmic scale. Solid lines denote the no-timing benchmark and the parent overlay; dashed lines denote the preregistered Line-A variants. Legends report Sharpe ratio, annualized return, and maximum drawdown computed from the frozen NAV files. In all three panels, no re-measurement of the morning cross-section improves on the parent overlay.

information invisible to any endpoint statistic. *Method:* slopes, acceleration, and curvature were constructed from the 09:31/09:45/09:55 path, subject to a mandatory orthogonality pre-screen; any significant composite was decomposed into its primitive slices. *Verdict:* *null, and the most instructive of the three.* A3 passed the orthogonality screen (correlation with level +0.14; with the parent signal +0.045), and the acceleration feature produced an in-sample $t = -4.45$ under double controls. Decomposition attributed the entire effect to the earliest slice (open→09:31), i.e. to the timing of the level move rather than to shape, and the one-shot OOS evaluation rejected it by sign flip ($t = +0.45$). Pure curvature was insignificant throughout ($t = -0.09$); strategy overlays were negative in all six cells (Figure 1, right panel).

3.2 Line B (outward): transplanting the recipe

B1 Tail-of-day breadth. *Motivation:* the recipe was mirrored onto the final compression window of the day (14:45–14:55, tradable at the close) to predict the overnight gap and the following day. *Method:* a new 10.8-million-row tail-snapshot cache was built and row-reconciled against the parent cache; IC tests, orthogonality checks against the parent, and close-to-close overlays at three cost tiers were conducted. *Verdict:* *proposition confirmed; no deployable alpha.* Tail breadth predicts the overnight gap (full-sample NW $t = 3.01$) and is nearly orthogonal to the parent (Pearson 0.106; signal-level $\phi = 0.072$). The central premise of the project, namely that an orthogonal information window exists, is therefore supported. The payoff, however, is concentrated in the overnight leg, which is structurally negative in A-shares (approximately -25% per year equal-weighted; Figure 2b), decays out of sample, concentrates in small-cap microstructure, and cannot absorb daily round-trip costs (Figure 2a). The signal reduces the losses of an unattractive leg; it does not generate returns.

B3 Post-holiday reopenings. *Motivation:* multi-day closures accumulate information, so the reopening morning should constitute the strongest compression window available to the mechanism. *Method:* holiday first days were tagged from calendar gaps ($n = 69$, of which 22 follow long holidays); interaction regressions were estimated with permutation inference in view of the small sample, and an execution-honest strategy variant differing from the parent only on reopening mornings was evaluated. *Verdict:* *null, and doubly informative.* The interaction is insignificant (full-sample -168 bps; permutation $p = 0.15$), attenuates five-fold from IS to OOS, and, most decisively, the mechanism gradient is reversed: the longest closures (gap ≥ 8 days), where accumulated information

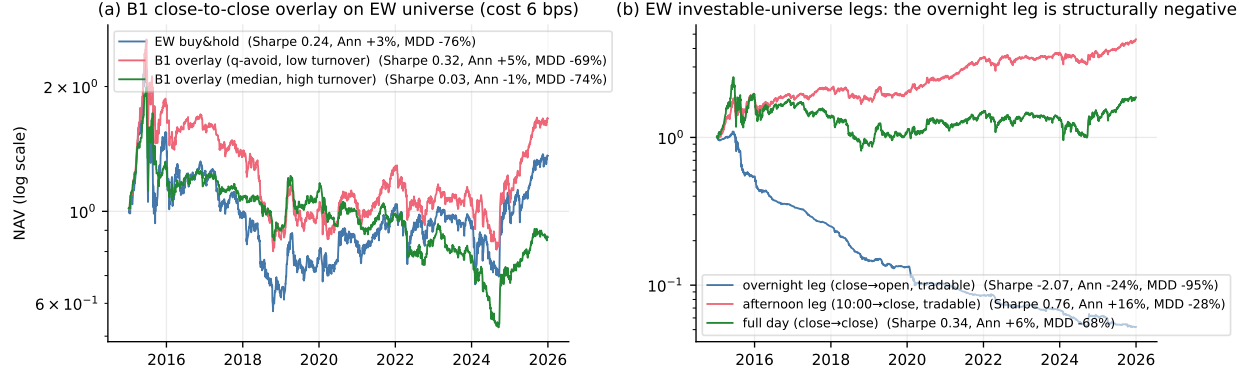


Figure 2: B1. (a) Close-to-close overlays on the equal-weighted investable universe at 6 bps costs: the low-turnover variant adds marginal value at low cost tiers only, and the median variant is eliminated by turnover. (b) Cumulative equal-weighted legs computed from the frozen B5 segment file: the tradable overnight leg (close→open) loses approximately 24% per year, while the afternoon leg carries a positive premium. (The estimate from B1’s own tail cache is -26.7% per year; the two files are consistent.) A signal whose payoff accrues overnight must first overcome this structural deficit.

is largest, show no effect (-26 bps, $p = 0.45$), with the entire point estimate loaded on short holidays (Figure 3a). The strategy variant differs from the parent on only 19 days in eleven years, and its Δ Sharpe is negative in every segment (Figure 3b). The holdings correlation with the parent is 0.98, so the variant is a regime-restricted modification rather than an independent factor. As a by-product, the evidence suggests that overnight-gap compression is already saturated by a single normal night, with additional closure time offset by reopening overreaction.

B5 Volatility timing. *Motivation:* the preceding directions targeted the first moment; the A1 placebo indicated that dispersion behaves as a volatility proxy, motivating a second-moment target, for which exposure scaling (rather than shorting) guarantees executability. *Method:* realized-variance targets were constructed from intraday segments. Candidate features were required to add value on top of a HAR baseline, since the persistence of volatility is the dominant confounder; forecast comparisons used Diebold–Mariano tests on out-of-sample QLIKE, and vol-targeting NAVs were compared against two baselines (constant exposure; HAR-only scaling). As a preliminary step, the residual value of B1 was monetization-tested as an overnight-avoidance overlay on the actual small-cap sleeve; the OOS Δ Sharpe was negative at all three cost tiers (Figure 4b). *Verdict:* null as a strategy, with the cleanest lesson of the project. Morning breadth genuinely predicts same-day remaining variance beyond HAR (increment $t = 6-9$; OOS DM -4.16 ; the only statistic in the project to survive Bonferroni $\times 25$). Vol-targeting with the enriched forecast is nevertheless indistinguishable from HAR-only scaling (Δ Sharpe ≈ 0 ; Figure 4a), and both variants underperform constant exposure (-0.12 full sample, -0.22 OOS): the afternoon leg carries a positive premium, high-volatility days are not low-return days, and de-risking them reduces the Sharpe ratio. Statistical significance did not translate into decision value.

4 Lessons and a reusable falsification protocol

The findings that survive the six rejections are collected here.

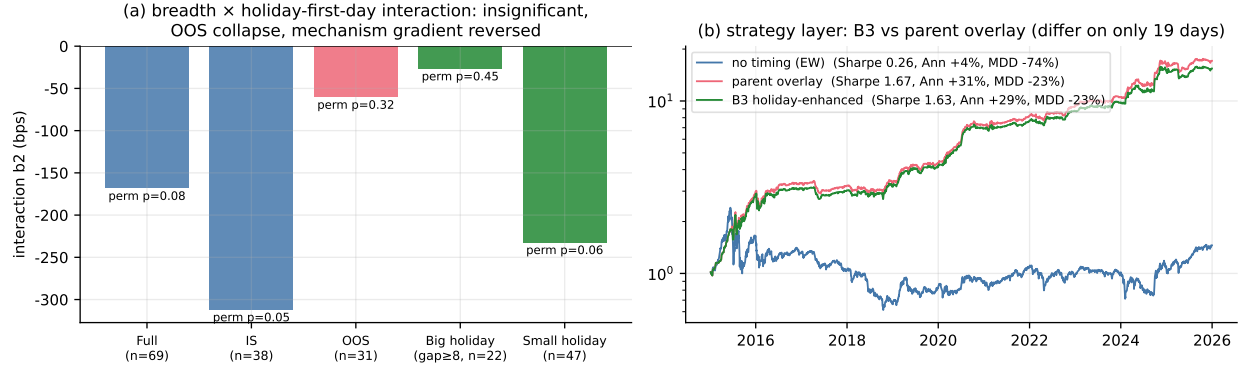


Figure 3: B3. (a) The breadth $\times$ holiday-first-day interaction coefficient with one-sided permutation p -values (5,000 label shuffles): insignificant overall, an IS-to-OOS attenuation, and a reversed dose-response in which the theoretically strongest regime (long holidays) shows no effect. (b) The execution-honest strategy layer: the holiday-enhanced variant differs from the parent on only 19 days and reduces the Sharpe ratio in all segments.

4.1 Two structural findings

(i) **The morning cross-section is informationally exhausted by raw breadth and level.** Line A approached the measurement side from three independent angles, dispersion (A1), economic weighting (A2), and temporal shape (A3), and failed for three distinct reasons: lack of orthogonality, lack of orthogonality, and orthogonality without content. Jointly, the three results indicate that no quantity computable from the 09:55 cross-section adds information to the two variables already used by the parent strategy.

(ii) **Orthogonal information exists but is not monetizable; existence and deployability are distinct claims.** B1 established the existence claim ($t = 3.01$, correlation 0.106). B1, B3, and B5 subsequently encountered three distinct obstacles between information and realized value: a payoff concentrated in an untradable leg (the overnight segment, structurally about -25% per year); collinearity with the parent (holdings correlation 0.98); and decision irrelevance (the HAR baseline already spans the exposure choice). Several features of A-share microstructure, including the negative overnight leg, the opening-auction artifact, short-sale constraints, and the positive afternoon premium, repeatedly blocked the final step toward implementability.

4.2 Six transferable principles

1. **Low correlation does not imply incremental value (A3).** Orthogonality is necessary but not sufficient: A3 was the most orthogonal candidate in the project and carried the least content.
2. **Statistical significance does not imply executable value (B5).** The evaluation chain should always be run to a net-of-cost Δ Sharpe against a baseline that already exploits lagged information; a regression t -statistic is not a stopping point.
3. **The tradable leg in which the payoff accrues should be identified early (B1).** A genuine signal whose payoff is confined to an untradable or structurally negative leg constitutes a finding, not a factor.
4. **The opening-price artifact must be excluded by construction** (encountered independently in B1 and B5). Equal-weighted open-to-close intraday legs can display returns of $+40\%$

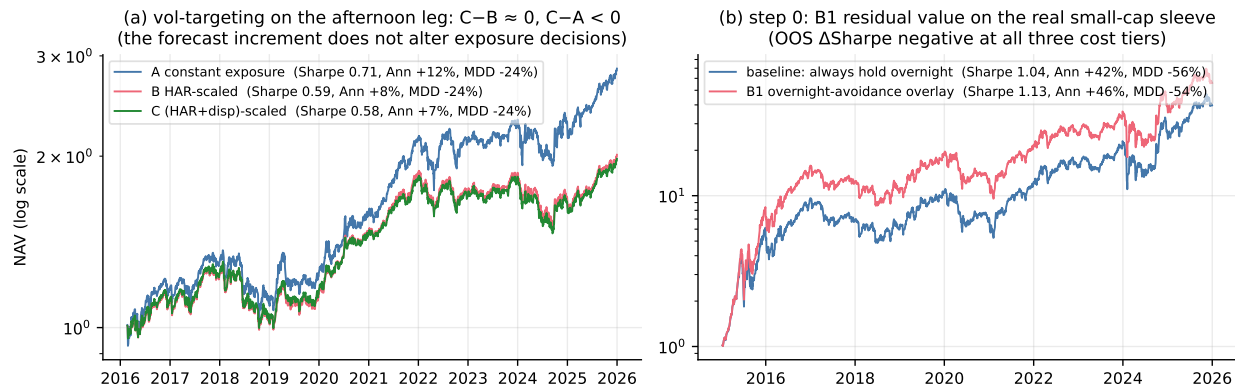


Figure 4: B5. (a) Vol-targeting NAVs on the afternoon leg (10:00→close; costs charged on exposure changes): the dispersion-enriched forecast (C) is indistinguishable from HAR-only scaling (B), and both trail constant exposure (A); a forecasting increment of $t = 6$ carries no decision value. (b) The preliminary step: B1's overnight-avoidance overlay on the actual small-cap sleeve is cosmetic in the full sample and negative out of sample at every cost tier.

per year attributable to bid–ask bounce; only close→open and close→close segments are admissible in performance accounting.

5. **Regime-restricted modifications of a parent strategy are not new factors (B3).** Holdings correlation exposes such variants directly (0.98 in this case), and calendar events add a small-sample hazard; permutation inference with reported effective n is required.
6. **The mechanism gradient should be tested, not only the pooled coefficient (B3).** Where theory predicts a dose–response (greater accumulation implying a stronger effect), monotonicity is testable, and a reversed gradient falsifies the mechanism even when the pooled estimate appears favorable.

4.3 A falsification toolkit

Table 2 lists the tests that produced the rejections in this study, in the order in which they should be applied. Each is inexpensive, generic, and suitable for encapsulation as a library function. Together they define a pipeline in which a candidate factor must survive every stage before any capital-relevant claim is made.

4.4 Intercepted false positives and test attrition

Three estimates in this study were individually strong enough to be reported as discoveries. All three were rejected by the protocol (Table 3); the project-level attrition is shown in Figure 5.

4.5 Process: two-layer verification and the signed ledger

Two process-level arrangements contributed as much as any individual statistic and are recommended for institutionalization.

Two-layer verification. Researchers were required to freeze every reported number to interpreter-produced files and to attempt self-falsification (decompositions, placebos, artifact checks). Acceptance was then conditional on independent re-execution of the headline numbers from the frozen

Table 2: The falsification toolkit. Tests are applied top to bottom; failure at any stage terminates the candidate.

Test	Question addressed	Rejection criterion	Rejections here
1. Level-orthogonality screen	Is the “new” measure the index move in disguise?	$ \text{corr} $ with level or raw baseline > 0.5	A2 (0.52); A1 in part
2. Nested HAC increment	Does the candidate add to the baseline in sample?	increment $t < 2$ with baseline present	A1, A2
3. Absorption regression	In a joint regression, which variable retains its t ?	candidate’s t absorbed by baseline	A2 (−4.08 vs +1.39)
4. Decomposition	Which primitive within a significant composite carries the effect?	driver is a disguised baseline slice	A3 (early-level slice)
5. One-shot OOS, sign-flip rule	Does the effect persist outside the fitting window?	sign flip or t collapse; OOS is spent once	A3 (−4.45 → +0.45), B3
6. Permutation and gradient tests (small n)	Could $n = 22$ –69 events produce the estimate by chance? Is the dose–response monotone?	permutation p large; gradient reversed	B3
7. Persistence baseline (HAR)	Is apparent volatility predictability mere autocorrelation?	no DM-significant OOS gain over HAR	B5 next-day RV
8. Run-to-net-value	Does the statistical edge alter decisions after costs?	$\Delta\text{Sharpe} \approx 0$ vs lagged-information baseline	B5 ($t = 6 \rightarrow \Delta \approx 0$)
9. Artifact registry	Does any leg rely on untradable prints (open→close EW, overnight, same-bar)?	payoff concentrated in an excluded leg	B1; B5 step 0
10. Placebo separation	Directional signal or volatility proxy?	symmetric amplification on placebo days	A1
11. Holdings correlation	Is the “new factor” a re-parameterized parent?	position correlation > 0.9	B3 (0.98)
12. Trial ledger with Bonferroni	Does any statistic survive the family-wide threshold?	$p >$ family threshold (0.05/25)	all but one statistic

artifacts. All three entries in the interception log were flagged by the originating researcher and confirmed by independent re-execution; no conclusion rests on unverified reporting.

Signed, append-only handoff ledger. Each stage appended a signed record with three mandatory boundary lines (verified, with evidence; assumed; unverified risks), together with a distilled lessons layer required reading for subsequent directions. The failure mode of A2 was anticipated by the lesson recorded for A1; the orthogonality pre-screen in A3 and the HAR baseline in B5 were both mandated by earlier entries. The ledger also carries the cumulative trial count on which family-wide deflation depends.

Two-layer validity. Every conclusion was stated at two levels: internal validity (whether the in-sample result is genuine, supported by engine self-checks reconciling the rebuilt signal exactly with the parent and by row-count reconciliation of caches) and external validity (whether it generalizes, supported by a wide survivorship-free cross-section, three sleeves, and one-shot OOS evaluation). First-order sample representativeness was treated as prior to second-order overfitting defenses.

Table 3: Interception log: apparently significant results and the tests that rejected them.

Apparent result	Origin	Rejected by	Lesson
acceleration $t = -4.45$ (IS, double-controlled)	A3 path shape	decomposition traced the effect to an early level slice; OOS sign-flipped to +0.45	in-sample significance can be a level-timing artifact
+41.7%/yr EW intra-day leg	B1 leg decomposition	identified as opening-auction bid-ask bounce; excluded from all performance figures; misleading files removed	untradable prints are not returns
HAR increment $t = 6-9$, survives Bonferroni $\times 25$	B5 remaining-variance forecast	vol-targeting Δ Sharpe against HAR-only scaling ≈ 0 ; constant exposure superior in either case	significance without decision value is not alpha

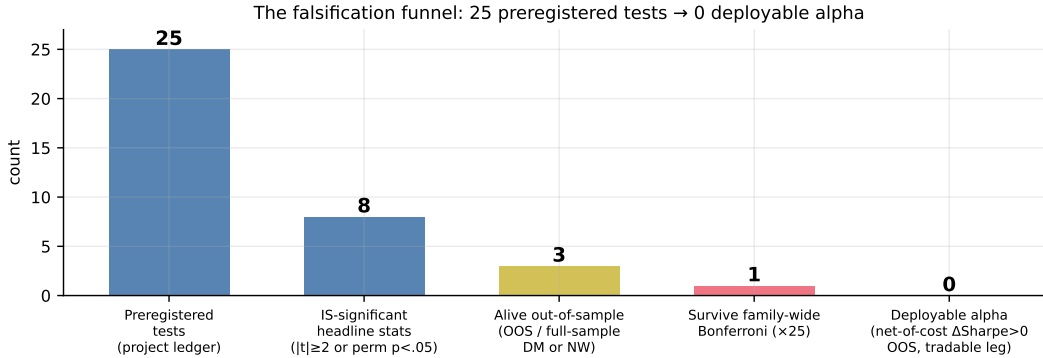


Figure 5: Test attrition. Stage definitions: 25 preregistered tests recorded in the project trial ledger; 8 headline statistics reached $|t| \geq 2$ (or permutation $p < 0.05$) in sample; 3 remained significant out of sample (the B1 full-sample IC and the B5 Diebold–Mariano tests for dispersion and breadth on remaining variance); 1 survived the family-wide Bonferroni threshold (breadth $\rightarrow$ same-day remaining variance, $p = 0.0008$); 0 converted into a positive net-of-cost, tradable-leg Δ Sharpe out of sample.

5 Conclusion

Six mechanism-motivated extensions of the validated early-session sentiment factor, comprising three inward re-measurements and three outward transplantations, were preregistered, executed under an execution-honest protocol, and rejected as sources of deployable alpha (25 trials; family-wide Bonferroni; no economic survivor). The study nevertheless returns value of three kinds: negative knowledge, in the form of six documented dead ends with identified failure modes; one confirmed proposition, namely that an orthogonal information window exists at the tail of the trading day, with monetization blocked by A-share microstructure; and a reusable falsification protocol (Table 2) whose twelve stages intercepted three apparent discoveries.

Should this line of research be revisited, the remaining mechanistic room lies in information sources other than price and volume (order-flow, disclosure events, margin-trading data) rather than in further transformations of morning price data. Any such study should be preregistered, carry its own out-of-sample period, and inherit the trial count established here.

A The seven honesty gates

Each direction was required to state, for every claim, which of the following gates had been passed.

1. **Point-in-time, no lookahead:** features dated $\leq t$, returns from t^+ ; same-bar lookahead explicitly checked; survivorship-free including delistings.
2. **Execution honesty:** fills at actual decision times (10:00 repricing, closing auction); excluded legs never enter performance accounting.
3. **Fair benchmarking:** identical sleeve, costs, and accounting on both sides; only same-basis differences are reported.
4. **Out-of-sample:** IS for direction and thresholds; OOS evaluated once and never iterated.
5. **Multi-tier costs:** 3/6/10 bps stress tiers applied to every positive claim.
6. **Multiple-testing deflation:** project-level trial ledger; family-wide Bonferroni; preregistered variants only.
7. **Executability:** no short positions assumed; capacity and leg tradability assessed.

B Trial accounting

A1: 4 (two measures, two overlay variants). A2: 4 (two weightings, two overlays). A3: 6 (three shape families, decomposition, two overlays). B1: 3 (two breadth measures with targets collapsed by preregistration, one overlay family). B3: 6 (pooled, IS, OOS, two gap tiers, one strategy variant). B5: 2 (two targets; the preliminary step inherits the B1 count). **Total: 25.** The family-wide Bonferroni threshold at $\alpha = 0.05$ is $p < 0.002$. The sole statistical survivor, morning breadth predicting same-day remaining variance ($p = 0.0008$), fails the run-to-net-value stage and is archived as genuine but not monetizable.

All numbers in this report were computed by scripts from frozen data files and independently re-executed before acceptance. The full audit trail, comprising the signed ledger, per-direction reports, frozen caches, and figures, resides in the project repository.